

CHE 312 HW# 3

Monday, September 15, 2025

9:24 AM

A.1) BALANCE EQUATIONS:

$$\frac{d^2 T_1(x)}{dx^2} = -\frac{\dot{q}_0(1+ax)}{k_1}, \quad \frac{d^2 T_2(x)}{dx^2} = -\frac{\hat{q}}{k_2}, \quad \frac{d^2 T_3(x)}{dx^2} = -\frac{\dot{q}_0(1+ax)}{k_3}$$

BOUNDARY CONDITIONS:

B.C.1: $T_1(x=0) = T_{s,1}$

B.C.2: $T_3(x=L_3) = T_{s,2}$

B.C.3: $k_1 \left. \frac{dT_1(x)}{dx} \right|_{x=L_1} = k_2 \left. \frac{dT_2(x)}{dx} \right|_{x=L_1}$

B.C.4: $k_2 \left. \frac{dT_2(x)}{dx} \right|_{x=L_2} = k_3 \left. \frac{dT_3(x)}{dx} \right|_{x=L_2}$

B.C.5: $T_1(x=L_1) = T_2(x=L_1)$

B.C.6: $T_2(x=L_2) = T_3(x=L_2)$

A.2) REGION 1:

$$\int \frac{d^2 T_1(x)}{dx^2} = \int -\frac{\dot{q}_0(1+ax)}{k_1}$$

$$\int \frac{dT_1(x)}{dx} = \int -\frac{\dot{q}_0}{k_1} \left(x + \frac{ax^2}{2} \right) + C_1$$

$$T_1(x) = -\frac{\dot{q}_0}{k_1} \left(\frac{x^2}{2} + \frac{ax^3}{6} \right) + C_1 x + C_2$$

if $q_1''(x) = -k_1 \frac{dT_1(x)}{dx}$,

$$q_1''(x) = \dot{q}_0 \left(x + \frac{ax^2}{2} \right) - k_1 C_1$$

REGION 2:

$$\int \frac{d^2 T_2(x)}{dx^2} = \int -\frac{\hat{q}}{k_2}$$

$$\int \frac{d^2 T_2(x)}{dx^2} = -\frac{\dot{q}}{K_2}$$

$$\int \frac{dT_2(x)}{dx} = -\frac{\hat{q}_0}{K_2} x + C_3$$

$$T_2(x) = -\frac{\hat{q}_0}{2K_2} x^2 + C_3 x + C_4$$

$$q_2''(x) = \hat{q}_0 - K_2 C_3$$

REGION 3:

$$\int \frac{d^2 T_3(x)}{dx^2} = -\frac{\dot{q}_0(1+ax)}{K_3}$$

$$\int \frac{dT_3(x)}{dx} = -\frac{\dot{q}_0}{K_3} \left(x + \frac{ax^2}{2} \right) + C_5$$

$$T_3(x) = -\frac{\dot{q}_0}{K_3} \left(\frac{x^2}{2} + \frac{ax^3}{6} \right) + C_5 x + C_6$$

$$q_3''(x) = \dot{q}_0 \left(x + \frac{ax^2}{2} \right) - K_3 C_5$$

B.1) left steel bar: $0 \leq x \leq b$

fuel: $b \leq x \leq b+2L$

right steel bar: $b+2L \leq x \leq 2b+2L$

BALANCE EQUATIONS:

$$K_f \frac{dT_f(x)}{dx} = -\dot{q}, \quad K_s \frac{dT_s(x)}{dx} = 0$$

BOUNDARY CONDITIONS:

$$\text{B.C.1: } \left. \frac{dT_s(x)}{dx} \right|_{x=0} = 0$$

$$\text{B.C.2: } -K_s \left. \frac{dT_s(x)}{dx} \right|_{x=b} = -K_f \left. \frac{dT_f(x)}{dx} \right|_{x=b}$$

$$\text{B.C.3: } -K_s \left. \frac{dT_s(x)}{dx} \right|_{x=b+2L} = -K_f \left. \frac{dT_f(x)}{dx} \right|_{x=b+2L}$$

$$\text{B.C.3: } -k_f \left. \frac{dT_f(x)}{dx} \right|_{x=b+2L} = -k_s \left. \frac{dT_s(x)}{dx} \right|_{x=b+2L}$$

$$\text{B.C.4: } -k_s \left. \frac{dT_s(x)}{dx} \right|_{x=2b+2L} = h \left[T_s(x=2b+2L) - T_\infty \right]$$

$$\text{B.C.5: } T_s(x=b) = T_f(x=b)$$

$$\text{B.C.6: } T_s(x=b+2L) = T_f(x=b+2L)$$

$$\int \frac{d^2 T_f(x)}{dx^2} = -\frac{\dot{q}}{k_f}$$

$$\int \frac{dT_f(x)}{dx} = -\frac{\dot{q}}{k_f} x + C_1$$

$$T_f(x) = -\frac{\dot{q}}{k_f} \frac{x^2}{2} + C_1 x + C_2$$

From B.C.1, left steel $0 \leq x \leq b$ is isothermal. Call temp T_0 .

$$\Rightarrow \left. \frac{dT_s(x)}{dx} \right|_{x=0} = \left. \frac{dT_f(x)}{dx} \right|_{x=b} = 0$$

$$C_2 = T_0$$

$$T_f(x-b) = -\frac{\dot{q}}{k_f} \frac{(x-b)^2}{2} + C_1(x-b) + C_2$$

$$\Rightarrow 0 = -\frac{\dot{q}}{k_f} b + C_1 \Rightarrow C_1 = \frac{\dot{q}}{k_f} b$$

$$T_f(x) = -\frac{\dot{q}}{k_f} \frac{x^2}{2} + \frac{\dot{q}}{k_f} x b + T_0$$

Now, for $T_s(x)$:

$$T_s(x) = T_0, \quad 0 \leq x \leq b$$

$$-k_f \left. \frac{dT_f(x)}{dx} \right|_{x=b+2L} = -k_s \left. \frac{dT_s(x)}{dx} \right|_{x=b+2L}$$

$$-k_f \left(-\frac{\dot{q}}{k_f} (b+2L) + \frac{\dot{q}}{k_f} b \right) = -k_s \left. \frac{dT_s(x)}{dx} \right|_{x=b+2L}$$

$$\dot{q} + 2\dot{q}L - \dot{q} = -k_s \left. \frac{dT_s(x)}{dx} \right|_{x=b+2L}$$

$$\left. \frac{dT_s(x)}{dx} \right|_{x=b+2L} = -\frac{2\dot{q}L}{k_s} \leftarrow \text{constant throughout}$$

$$\left. \frac{dT_s(x)}{dx} \right|_{x=b+2L} = -\frac{2\dot{q}L}{K_s} \leftarrow \text{constant throughout}$$

$$-K_s \left. \frac{dT_s(x)}{dx} \right|_{x=2b+2L} = h [T_s(x=2b+2L) - T_\infty]$$

$$-\cancel{K_s} \left(-\frac{2\dot{q}L}{\cancel{K_s}} \right) = h [T_s(x=2b+2L) - T_\infty]$$

$$\frac{2\dot{q}L}{h} = T_s(x=2b+2L) - T_\infty$$

$$T_s(x=2b+2L) = T_\infty + \frac{2\dot{q}L}{h}$$

$$\Rightarrow T_s(x) = T_\infty + \frac{2\dot{q}L}{h} + \frac{2\dot{q}L}{K_s} (2b+2L-x)$$

Use B.C.3 to find T_0

$$T_f(b+2L) = T_0 - \frac{2\dot{q}L^2}{K_f}$$

$$T_s(b+2L) = T_\infty + \frac{2\dot{q}L}{h} + \frac{2\dot{q}L}{K_s} (2b+2L-b-2L)$$

$$T_0 - \frac{2\dot{q}L^2}{K_f} = T_\infty + \frac{2\dot{q}L}{h} + \frac{2\dot{q}Lb}{K_s}$$

$$T_0 = T_\infty + \frac{2\dot{q}L}{h} + \frac{2\dot{q}Lb}{K_s} + \frac{2\dot{q}L^2}{K_f}$$

$$T_f(x) = -\frac{\dot{q}}{K_f} \frac{x^2}{2} + \frac{\dot{q}}{K_f} xb + T_\infty + \frac{2\dot{q}L}{h} + \frac{2\dot{q}Lb}{K_s} + \frac{2\dot{q}L^2}{K_f}$$

$$q_f''(x) = -\frac{\dot{q}}{K_f} x + \frac{\dot{q}}{K_f} b$$

$$T_s(x) = T_\infty + \frac{2\dot{q}L}{h} + \frac{2\dot{q}L}{K_s} (2b+2L-x)$$

$$q_s''(x) = -\frac{2\dot{q}L}{K_s}$$

B.2)

